

Hypothesis testing

→ It is a statistical method used to decide whether there's enough evidence in a sample to support a claim about a population.

→ We always has two statements :-

① Null hypothesis → Nothing special is happening.

② Alternative hypothesis → Something is happening.

→ Ex: ①

you created a new landing page & conversion rate looks higher.

old conversions = 5%.

New conversion = 7%.

Q → Is this improvement real or just random chance?

→ Step-1 :- Define Hypotheses.

H_0 (Null) :- New page has no effect (conversion = 5%).

H_1 (Alternative)

→ New page improves conversion (conversion > 5%)

→ Step-2 :- Col
you run A/B

you record con

→ Step-3 :- C
→ Software

→ Suppose

→ Step-4 :-

If P-Val

Here :- 0

→ Conclusion

improves

Not luc

→ P-Value

H_0 were

So, S

La

Step-2 :- Collect Data

you run A/B test :- (1) 1000 users See old pages
(2) 100 users See new pages

you record Conversations.

Step-3 :- Calculate Test Statistics + p-value.

Software does this.

Suppose you get : $p\text{-value} = 0.02$.

Step-4 :- Decision

If $P\text{-Value} < 0.05 \rightarrow \text{reject } H_0$

Here :- $0.02 < 0.05 \rightarrow \text{Reject } H_0$

→ Conclusion :- The new landing page really improves conversion.

Not luck, Not Coincidence, Real effect.

→ P-Value :- probability of getting this result if H_0 were true.

So, Small $P\text{-value} \rightarrow H_0$ is unlikely $\rightarrow$ reject it.

large $P\text{-value} \rightarrow$ data is normal $\rightarrow$ don't reject H_0

Type 1 & Type 2 errors :-

→ Type-1 error :- Rejecting H_0 when H_0 is actually true.

→ "You think something happened... but it didn't".

Ex:- A company tests a new ad campaign.

H_0 → New ad does not increase sales.

H_1 → New ad increase sales.

→ Data shows improvement → you reject H_0 → launch the ad.

→ But in reality... sales increase was just random luck.

→ you made a type 1 error.

"you celebrated. you were wrong".

→ [Type 1 = False Alarm.]

→ Type 2 error :- Failing to reject H_0 when H_0 is actually false.

"Something real is happening... but miss it".

→ Same example :- Same ad campaign.

Data shows small improvement → you keep H_0 → do nothing.

→ But in reality the ad really does increase sales.

→ you made a type 2 error.

→ you missed a good opportunity.

[Type 2 = Missed opportunity].

One tail & Two tail test :-

→ first the big idea :- This depends on what your question

is.

→ "Do I care about change in one direction or both?"

→ one tail test :- A one tailed test checks for an effect in only one direction.

→ you are testing → "Is it greater than?"

(or)

"Is it less than?"

Not both.

is

Ex:- "My new study method increases average marks above 70."

→ So you only care about: Higher than 70. not lower.

→ Hypotheses :-

H_0 : mean = 70

H_1 : mean > 70 (one direction)

This is right tailed test.

→ Interpretation :-

- You are only interested in improvement.
- If marks decrease, you don't even care. (for this case)

→ Left Tailed test :- If you say

"This medicine reduce blood pressure below 120"

$$H_0 \rightarrow BP = 120$$

$$H_1 \rightarrow BP < 120$$

This is a left tailed test.

→ "One direction, one arrow, one focus".

→ Two tailed test

A two tailed test checks for an effect in both directions.

→ "Is it different?"

"Not better" or worse. just different.

Ex:- This new algorithm changes prediction accuracy. you did not say increase or decrease. just change.

→ Hypotheses :-

$$H_0 \rightarrow \text{Accuracy} = \text{old accuracy}$$

$$H_1 \rightarrow \text{Accuracy} \neq \text{old accuracy} \text{ (both sides)}$$

Interception :- If accuracy increases $\rightarrow$ Significant.
If accuracy decreases $\rightarrow$ Significant.

